



Fluid Flow Through Fluidized Bed

Group #2
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Introduction

- **Purpose of lab**
 - To determine the superficial gas velocity needed for fluidization
 - To study the correlation between pressure drop and the flowrate of the fluid being passed through the fluidized bed

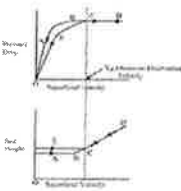


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Theory

- The relationship between pressure drop across the bed and the superficial velocity of the gas can be seen in figure 1, where the arrows represent increasing and decreasing gas velocity [3].
- Decreasing the velocity, the gas does not have the same coefficient of friction to overcome resulting in a different pressure drop curve.




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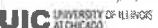


Theory continued...

- At low gas velocities the pressure drop through the fluidized bed can be described by the Ergun equation seen below:

$$\frac{\Delta P}{L} = \frac{150 \bar{V}_0 \mu (1-\epsilon)^2}{\phi_s^2 D_p^2 \epsilon^3} + \frac{1.75 \rho \bar{V}_0^2 (1-\epsilon)}{\phi_s D_p \epsilon^3} \quad (1)$$

ΔP = pressure drop across the bed
 L = bed height
 ρ = fluid density
 $\bar{V}_0$ = superficial velocity (based on empty cross section of bed)
 ϵ = bed voidage at minimum fluidization
 μ = fluid viscosity
 ϕ_s = sphericity
 D_p = particle diameter



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DL2 Type instead of pictures.
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Theory continued...

- Semi-empirical formulas have been developed for calculating minimum fluidization velocity based on particles Reynolds numbers

$$\bullet \frac{\Delta P}{L} = g(1 - \epsilon_M)(\rho_p - \rho) \quad (2)$$

ϵ_M = bed voidage at minimum fluidization
 ρ_p = particle density

Theory continued...

- Combining the Ergun equation (1) and the semi-empirical equation (2) the minimum velocity can be described by equation (3).

$$\bullet g(\rho_p - \rho) = \frac{150 \dot{V}_{0M}}{\phi_s^2 D_p^2} \frac{(1 - \epsilon_M)}{\epsilon_M^3} + \frac{1.75 \rho \dot{V}_{0M}^2}{\phi_s D_p} \frac{1}{\epsilon_M^3} \quad (3)$$

Theory continued...

- For very small particles, the equation for minimum velocity becomes equation (4).

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$$\bullet \dot{V}_{0M} \approx \frac{g(\rho_p - \rho)}{150 \mu} \frac{\epsilon_M^3}{(1 - \epsilon_M)} \phi_s^2 D_p^2 \quad (4)$$

Theory continued...

- When calculating the average particle diameters equation (5) is implemented.

$$\bullet D_{pAV} = \frac{1}{\sum (x_i / D_{pi})} \quad (5)$$

D_{pAV} = average particle diameter

x_i = mass fraction of particle size D_{pi} (where D_{pi} is taken from the sieve size)

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DL6 Show the postulates getting equation (4).

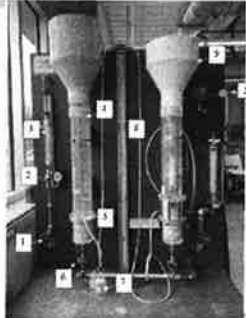
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Apparatus

Number	Materials	Brief Description
1	Air Flow Valve	Controls the flow rate of air
2	Pressure Gauge	Measures the pressure of air
3	Rotameters	Reads the Pressure of the Flow of Air
4	Fluidized Bed Vessel	Supports the Resins and the Sand
5	Air Distributing Plate and Beads	Distributes the air through the column to have even surface area
6	Ball Valve	Controls the flow of air to the vessel
7	Trap	Captures excess material before entering the manometer
8	Manometer	Reads the pressure differential between the bottom and the top of vessel
9	Disengaging Section	Used to allow particles to remain within the vessel and allows air to leave



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Material & Supplies

Materials	Brief Description
Air	Used to pump air through the column
Sand	Used for the Bed for the Sand
Ion Exchange Resin	Used for the Bed for the Ion Exchange Resin
Ladder	Used to climb to add the material into the fluidized bed vessels
Vacuums & Hoses	To clean the vessels to take out the materials

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Procedure

Fine Silica Sand Bed Material

- Pack vessel to 6" height
- Record pressure drop as a function of gas flowrate by changing the flow of air slowly
 - Increase air flowrate until pressure drop becomes constant and record pressure drops throughout
 - Decrease flowrate and record pressure drops as flowrate brought down to zero.

- Repeat the experiment for a 10" packed bed

Ion Exchange Resin Beads

- Same procedure as above is done for this material

Calculate void fraction by adding 100 mL of water and 100 mL of resin in a graduated cylinder and measure how much the total volume is.

For size of the particle choose 5 different sieve trays $\frac{75\mu}{10}$ add a weighed amount of resin on the sieves and weigh each tray individually.

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Experimental Challenges

- Have one person read the manometer so it is read the same way each time; though it can become troublesome towards the end since the person is reading the measurements the whole-time during lab

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Safety

- Make sure to be careful with the ladder while filling and cleaning the fluidized bed vessel
- Wear nitrile gloves when dealing with the materials
- Know where the safety air relief valve is in case pressure goes above 40 psig
- Make sure put the vacuum hose in the vacuum hole, not the blower hole

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